

$$\underline{\eta_{TS} = 0.7} = \frac{1 - \frac{T_{02}}{T_{01}}}{1 - \left(\frac{p_2}{p_{01}}\right)^{\frac{\gamma-1}{\gamma}}} = \frac{1 - \frac{T_{02}}{T_{01}}}{1 - \left(\frac{0.14}{2}\right)^{\frac{1.3-1}{1.3}}}$$

$$\Rightarrow \frac{T_{02}}{T_{01}} = 0.6790$$

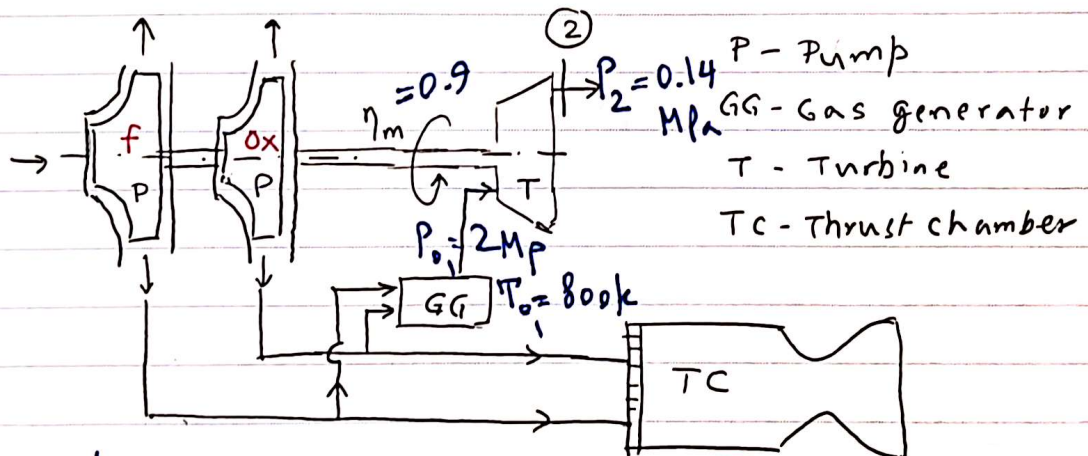
$$T_{02} = 0.6790 T_{01} = 0.6790 (800) = 543.2 \text{ K}$$

$$P_T = \dot{m}_T \underbrace{(1060)}_{c_p} \underbrace{(800 - 543.2)}_{\checkmark} = 272208 \dot{m}_T$$

$$\eta_m P_T = P_p \quad \checkmark \quad \checkmark$$

$$(0.90)(272208 \dot{m}_T) = \underbrace{6(10^5)}_{0.6 \text{ MW}} \Rightarrow \dot{m}_T = 2.45 \frac{\text{kg}}{\text{s}}$$

$$\text{Fraction} = \frac{\dot{m}_T}{\dot{m}} = \frac{2.45}{92} = 0.0266, \text{ or } 2.66\%$$



Solve $\eta_{TS} \rightarrow$ get
 $T_0 \rightarrow$ get w_T

$$\bar{M} = 34$$

$$\gamma = 1.3$$

$$\eta_{TS} = 0.7$$

get P_T then

$$P_T = \frac{P_p}{\eta_m}$$

Quiz 9

$$P_T = \dot{m}_T w_T = \dot{m}_T \eta_{TS} w_{TS} = \frac{\dot{m}_P w_P}{\eta_m}$$

unknown

key relation $\Rightarrow$

$$P_T = \frac{P_p}{\eta_m}$$

$$\text{Pressure head or head} = \frac{P}{\rho g} \quad \begin{matrix} \text{or} \\ \text{[m]} \\ \text{[ft]} \end{matrix}$$

Ex:

$$\text{pressure} = 101325 \text{ Pa}$$

$$\text{Water head} = \frac{101325}{(1000)(9.81)} = 10.33 \text{ m}$$

(SG = 1)

$$\text{Mercury head} = \frac{101325}{(13600)(9.81)} = 0.76 \text{ m} = 760 \text{ mm}$$

(SG = 13.6)

$$\text{Lox head} = \frac{101325}{(1140)(9.81)} = 9.06 \text{ m}$$

(SG = 1.140)

A pump becomes susceptible to cavitation if the fluid pressure is less than its vapor pressure.

$$P < P_v \rightarrow \text{Cavitation}$$

$$SG_{Lox} = 1.14, \dot{m} = 182 \text{ kg/s}, z_{loss} = 1 \text{ m}$$

$$p_{o1} = p_{tank} + \underbrace{\rho_{Lox} g z}_{=3} - \underbrace{\rho_{Lox} g z_{loss}}_{\text{friction}}$$

Divide by $\rho_{Lox} g$:

$$\frac{p_{o1}}{\rho_{Lox} g} = \frac{p_{tank}}{\rho_{Lox} g} + z - z_{loss}$$

$$z_{o1} = z_{tank} + z - z_{loss}$$

$$z_{tank} = \frac{241(1000)}{(1140)(9.81)} = 21.55 \text{ m}$$

$$z_{o1} = 21.55 + 3 - 1 = 23.55 \text{ m}$$

$$p_{v,Lox} = 101.3 \text{ kPa at } 90.3 \text{ K}$$

$$z_{v,Lox} = \frac{p_{v,Lox}}{\rho_{Lox} g} = \frac{101.3(1000)}{(1140)(9.81)} = 9.06 \text{ m}$$

$$\left(p_1 = p_{o1} - \rho_{Lox} \frac{u_1^2}{2} \right) \xrightarrow{\div \rho_{Lox} g} \left(z_1 = z_{o1} - \frac{u_1^2}{2g} \right)$$

$$p_o = p + \frac{1}{2} \rho u^2$$

Bernoulli Eq

$$u_1 = \frac{\dot{m}_1}{\rho_{Lox} A_1} = \frac{\dot{m}_1}{\rho_{Lox} \frac{\pi D_1^2}{4}} = \frac{182}{(1140)(0.009503)} = 16.8 \text{ m/s}$$

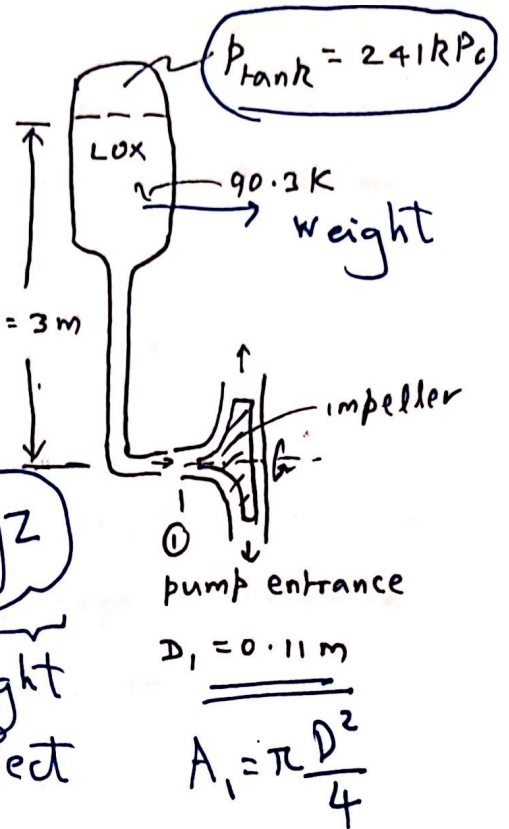
$$z_1 = \underbrace{z_{o1}}_{23.55} - \frac{u_1^2}{2g} = 23.55 - \frac{16.8^2}{2(9.81)} = 9.16 \text{ m}$$

$z_1 > z_{v,Lox}$, no cavitation

$z_1 > z_v \rightarrow$
no-cavitation

$$NPSH, \text{ see later, is } \frac{p_{o1} - p_v}{\rho_{Lox} g} = z_{o1} - z_v = 23.55 - 9.06 = 14.49 \text{ m}$$

This is $(NPSH)_A$. The value of $(NPSH)_p$ is experimentally deduced.



Cavitation is a significant limitation on pump performance. This phenomenon describes the formation of vapor bubbles in regions where the fluid static pressure declines to levels below the fluid's vapor pressure. Excessive bubble formation $P < P_v$ causes the flow rate to significantly ^{in declines} decline. Also, the ^{→ bubbles} appear collapse of these bubbles as they reach higher pressure regions ^{bubbles → destroy →} results in erosion of pump surfaces. Of the two, the latter is not considered a serious problem in rocket engines owing to their short duration of operation. In reusable engines (such as the SSME), however, the erosion effect does become important.

The variable $(P_{01} - P_v)$ is significant in analysis of cavitation.

The associated head is known as the net positive suction head (NPSH).

$$NPSH = \frac{P_{01} - P_v}{\rho g_0}$$

$\xrightarrow{\text{Total Pressure}} \quad \xrightarrow{\text{Vapour Pressure}}$

Here, P_{01} is the stagnation pressure of the fluid at the entrance to the pump (known as suction side) and P_v is the fluid vapor pressure.

The suction of the pump causes fluid to accelerate into it, and fall in pressure. If the acceleration is large enough, the static pressure reaches the vapor pressure value and

$$P \rightarrow P_v$$

cavitation becomes likely. As it happens, some cavitation is acceptable and performance problems arise only with excessive cavitation. Experiments show that a value for NPSH exists below which pump head declines dramatically. This value is called the required suction head $[(NPSH)_R]$ and is dependent on the fluid of interest. Thus, to avoid cavitation-related performance problems, the available NPSH $[(NPSH)_A]$ should be kept more than that required $[(NPSH)_R]$.

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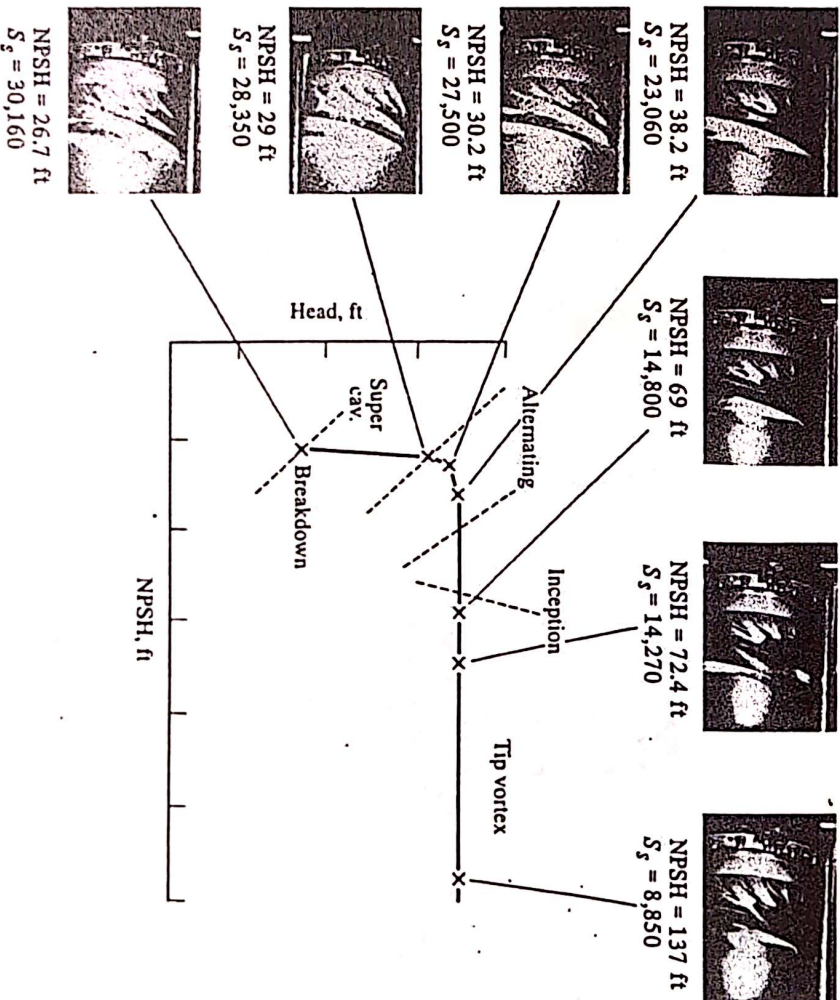


FIGURE 13.15 Experimental Inducer cavitation characteristics. (Courtesy Rockwell International, Rocketdyne Division.)

$$NPSH = \frac{P_0 - P_v}{\rho g}$$

Ω : rotational
speed of
the pump

$$S_g = \text{Suction Specific Speed} = \frac{\Omega \sqrt{Q}}{(g_e NPSH)^{3/4}}$$

SLANT ROCKETS

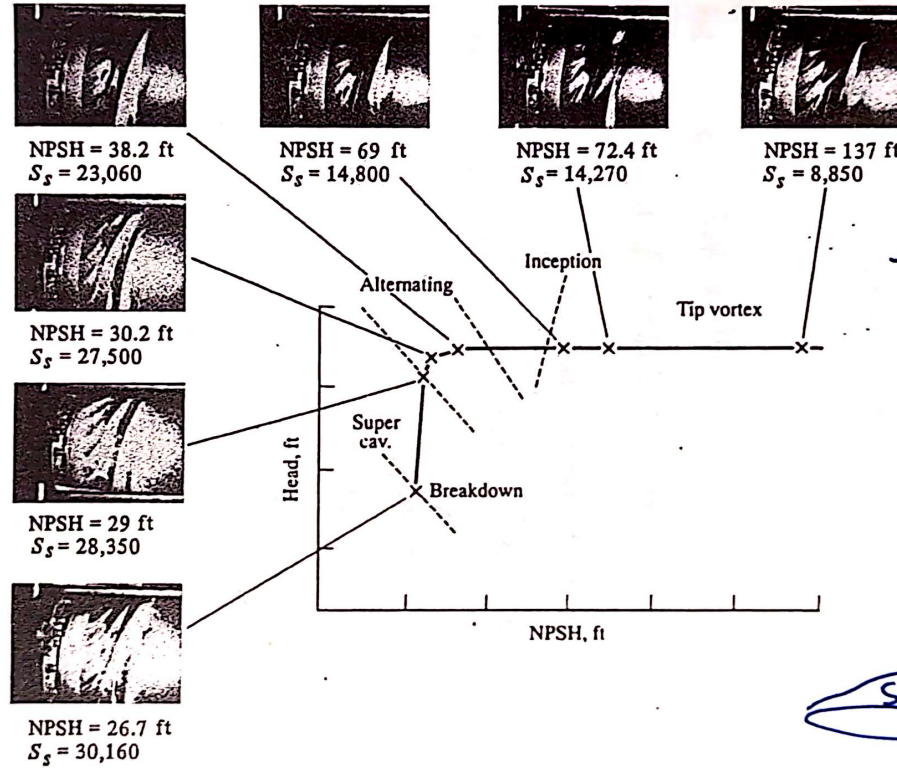
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$P < P_v$
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Cavitation

SuperCavitation

Super
 Cavitation

FIGURE 13.15 Experimental inducer cavitation characteristics. (Courtesy Rockwell International, Rocketdyne Division.)